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Areas of Expertise

Computer Vision, Deep Learning, Remote Sensing, Medical Imaging, Multimodal Integration

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1 Education

2005–2010	Ph.D. in Computer Science Adviser: Robert Pless, Ph.D. Thesis: Calibrating and Using the Global Network of Outdoor Webcams	Washington University in St. Louis
1995–1999	B.S. in Computer Science (Minor in Mathematics) <i>Summa Cum Laude</i> with Honors	University of Missouri

2 Appointments and Affiliations

Active

Vice Provost for Artificial Intelligence May 2026–	Washington University St. Louis, MO
Professor (with Tenure) 2022–	Dept. of Computer Science & Engineering, Washington University St. Louis, MO
Affiliated Faculty 2022–	Division of Computational & Data Sciences, Washington University St. Louis, MO

Affiliated Faculty 2022–	Imaging Science Program, Washington University <i>St. Louis, MO</i>
Affiliated Faculty 2023–	Electrical and Systems Engineering, Washington University <i>St. Louis, MO</i>
Affiliated Faculty 2023–	AI for Health Institute, Washington University <i>St. Louis, MO</i>
Associate Faculty 2022–	Taylor Geospatial Institute <i>St. Louis, MO</i>
Faculty Scholar 2023–	Center for the Environment, Washington University <i>St. Louis, MO</i>
Biodiversity Fellow 2022–	Living Earth Collaborative <i>St. Louis, MO</i>
Technical Consultant / Owner / Founder 2019–present	Multidomain Vision Research, LLC <i>St. Louis, MO</i>

Prior

Assistant Vice Provost for Digital Transformation Jan 2025–May 2026	Washington University <i>St. Louis, MO</i>
Professor (with Tenure) 2021–2022	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Director of Graduate Studies (Data Science) 2020–2022	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Member 2017–2022	Institute for Biomedical Informatics, University of Kentucky <i>Lexington, KY</i>
Associate Professor (with Tenure) 2016–2021	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Co-Department Chair (interim) 2019–2020	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Affiliated Faculty 2010–2019	Center for Visualization and Virtual Environments, University of Kentucky <i>Lexington, KY</i>
Visiting Research Scientist (sabbatical) 2017–2018	Orbital Insight, Inc. <i>Mountain View, CA</i>
Assistant Professor 2010–2016	Dept. of Computer Science, University of Kentucky <i>Lexington, KY</i>
Computer Vision Research Intern 2008	ObjectVideo, Inc. <i>Reston, VA</i>
Graduate Research Assistant 2005–2010	Dept. of Computer Science & Engineering, Washington University <i>St. Louis, MO</i>

3 Awards

- Best Paper Award (out of 75 valid submissions) [EarthVision Workshop 2024 at IEEE/CVF Computer Vision and Pattern Recognition (CVPR)]
- Highlighted Reviewer Recognition (top 8%) [ICLR 2022]
- Outstanding Reviewer Recognition [BMVC 2021]
- Outstanding Reviewer Recognition (top 10%) [NeurIPS 2020]
- Outstanding Reviewer Recognition [ICCV 2019]
- University of Kentucky, College of Engineering Dean’s Award for Excellence in Research [2018]
- Google Faculty Research Award [2018]
- Outstanding Reviewer Recognition [CVPR 2017]
- National Science Foundation CAREER Award [2016]
- Google Faculty Research Award [2016]
- Best Student Paper Award at Applied Imagery Pattern Recognition [2009]
- Ph.D. Forum Prize at the ACM/IEEE International Conference on Distributed Smart Cameras [2009]
- Best Talk Award for the Doctoral Student Seminar, Department of Computer Science, the Washington University in St. Louis, [Fall 2006]

4 Publications

Preprints

- [1] Y. Cui, F. Qiao, and N. Jacobs, *StereoGenBench: A synthetic multi-camera benchmark for stereo generation under controlled baseline regimes*, May 2026. arXiv: [2605.23237 \[cs.CV\]](#).
- [2] C. Robinson, G. Muhawenayo, S. Khanal, Z. Fang, I. Corley, A. M. Tárano, L. Estes, J. Marcus, N. Jacobs, H. Kerner, I. Becker-Reshef, and J. M. L. Ferres, *The first global agricultural field boundary map at 10m resolution*, May 2026. arXiv: [2605.11055 \[cs.CV\]](#).
- [3] H. J. Koch, J. Zhu, S. Bhatt, N. Jacobs, and J. M. Dortch, “Assessing deep learning segmentation models for mapping landslides from lidar-based elevation data,” Mar. 2026. DOI: [10.22541/essoar.177265426.62613754/v1](#) [Online]. Available: <http://dx.doi.org/10.22541/essoar.177265426.62613754/v1>
- [4] Y. Luo, F. Qiao, Z. Xiong, Y. Li, and N. Jacobs, *GenOpticalFlow: A generative approach to unsupervised optical flow learning*, Mar. 2026. arXiv: [2603.22270 \[cs.CV\]](#).
- [5] S. Sastry, D. Cher, B. Wei, A. Dhakal, S. Khanal, D. Gupta, and N. Jacobs, *TerraDiT: Point-conditioned diffusion transformer for satellite image synthesis*, Mar. 2026. arXiv: [2603.02172 \[cs.CV\]](#).
- [6] Z. Xiong, Y. Song, L. He, W. Xiong, Y. Yuan, F. Qiao, and N. Jacobs, *PhysAlign: Physics-coherent image-to-video generation through feature and 3d representation alignment*, Mar. 2026. arXiv: [2603.13770 \[cs.CV\]](#).
- [7] F. Qiao, Z. An, Z. Xiong, S. Belongie, and N. Jacobs, *Track2View: 4d-consistent camera-controlled video generation via paired 3d point tracks*, 2026. arXiv: [2606.15534 \[cs.CV\]](#).

- [8] Z. Xiong, Y. Song, H. Kang, Q. Yan, L. Jiang, J. Yang, Z. Fu, S. Fotiadis, A. Wang, Z. Liu, B. Liu, Y. Yang, X. Lu, and N. Jacobs, *ActWorld: From explorable to interactive world model via action-aware memory*, 2026. arXiv: [2606.17730 \[cs.CV\]](#).
- [9] C. Mosig, T. Kattenborn, D. Montero Loaiza, J. Vanja-Jehle, J. Brandt, N. Jacobs, S. Khanal, E. Xing, M. Schwartz, H. C. Muller-Landau, M. Beloiu, A. Bozzini, Y. Cheng, K. Ganz, B. Grüning, H. Hartmann, J. Hempel, S. Horion, S. Junttila, K. Korznikov, G. Kraemer, M. Mönks, D. Nardi, P. Neumeier, J. Schmid, S. Soltani, M. Therese-Schmehl, J. Veitch-Michaelis, and M. Mahecha, “Sub-pixel mapping of disturbance and tree mortality dynamics from sentinel-2 time series around the globe,” *Eartharxiv*, 2026. DOI: [10.31223/X5B18W](#) [Online]. Available: <https://eartharxiv.org/repository/view/11912/>
- [10] K. Klemmer, E. Rolf, M. Russwurm, G. Camps-Valls, M. Czerkawski, S. Ermon, A. Francis, N. Jacobs, H. R. Kerner, L. Mackey, G. Mai, O. M. Aodha, M. Reichstein, C. Robinson, D. Rolnick, E. Shelhamer, V. Sitzmann, D. Tuia, and X. Zhu, *Earth Embeddings: Towards AI-centric representations of our planet*, Dec. 2025. DOI: <https://doi.org/10.31223/X5HX9S>

Refereed Conference Papers

- [1] D. Cher, H. Iqbal, E. Xing, B. Wei, and N. Jacobs, “Tessellating the earth,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: [2606.27514 \[cs.CV\]](#).
- [2] T. Gao, J. Lin, D. Beaulieu, and N. Jacobs, “Trajectory-aware cross-view geo-localization with sequential observations,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: [2607.15491 \[cs.CV\]](#).
- [3] B. Wei, S. Sastry, D. Cher, E. Xing, and N. Jacobs, “TerraDiT-Ω: Unified spatial control for satellite image synthesis with any geospatial primitive,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. eprint: [2606.31029](#) (cs.CV).
- [4] Z. Xiong, X. Ye, B. Yaman, S. Cheng, Y. Lu, J. Luo, N. Jacobs, and L. Ren, “UniDrive-WM: Unified understanding, planning and generation world model for autonomous driving,” in *European Conference on Computer Vision (ECCV)*, Sep. 2026. arXiv: [2601.04453 \[cs.CV\]](#).
- [5] A. Dhakal, S. Khanal, S. Sastry, J. Arndt, P. A. Dias, D. Lunga, and N. Jacobs, “SimLBR: Learning to detect fake images by learning to detect real images,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026. arXiv: [2602.20412 \[cs.CV\]](#).
- [6] G. Muhawenayo, C. Robinson, S. Khanal, Z. Fang, I. Corley, A. Wollam, T. Gao, L. Strnad, R. Avery, L. Estes, A. M. Tárano, N. Jacobs, and H. Kerner, “PRUE: A practical recipe for field boundary segmentation at scale,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026.
- [7] S. Sastry, S. Khanal, A. Dhakal, J. Lin, D. Cher, P. Jarosz, and N. Jacobs, “ProM3E: Probabilistic masked multimodal embedding model for ecology,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2026. arXiv: [2511.02946 \[cs.CV\]](#).
- [8] A. Sarkar, S. Sastry, A. Pirinen, N. Jacobs, and Y. Vorobeychik, “DiffVAS: Diffusion-guided visual active search in partially observable environments,” in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, (oral), May 2026. arXiv: [2605.15519 \[cs.CV\]](#).
- [9] D. Cher, B. Wei, S. Sastry, and N. Jacobs, “VectorSynth: Fine-grained satellite image synthesis with structured semantics,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Mar. 2026. arXiv: [2511.07744 \[cs.CV\]](#).
- [10] A. Wollam, K. Ashley, M. Shugaev, O. Arend, I. Y. Semenov, H. Dashtestani, S. Ravi, and N. Jacobs, “Towards unconstrained cross-view pose estimation,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Mar. 2026.
- [11] A. Elallaf, N. Jacobs, X. Ye, M. Chen, and G. Liang, “Beta distribution learning for reliable roadway crash risk assessment,” in *Association for the Advancement of Artificial Intelligence (AAAI)*, Jan. 2026. arXiv: [2511.04886 \[cs.CV\]](#).

- [12] E. Xing, A. Stylianou, R. Pless, and N. Jacobs, “QuARI: Query adaptive retrieval improvement,” in *Neural Information Processing Systems (NeurIPS)*, vol. 2505.21647, Dec. 2025. arXiv: [2505.21647 \[cs.CV\]](#).
- [13] F. Qiao, Z. Xiong, E. Xing, and N. Jacobs, “Towards open-world generation of stereo images and unsupervised matching,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, Oct. 2025. arXiv: [2503.12720 \[cs.CV\]](#).
- [14] S. Sastry, A. Dhakal, E. Xing, S. Khanal, and N. Jacobs, “Global and local entailment learning for natural world imagery,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, vol. 2506.21476, Oct. 2025. arXiv: [2506.21476 \[cs.CV\]](#).
- [15] A. Dhakal, S. Sastry, S. Khanal, A. Ahmad, E. Xing, and N. Jacobs, “RANGE: Retrieval augmented neural fields for multi-resolution geo-embeddings,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025. arXiv: [2502.19781 \[cs.CV\]](#).
- [16] E. Xing, P. Kolouju, R. Pless, A. Stylianou, and N. Jacobs, “ConText-CIR: Learning from concepts in text for composed image retrieval,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025. arXiv: [2505.20764 \[cs.CV\]](#).
- [17] H. R. Kerner, S. Chaudhari, A. Ghosh, C. Robinson, A. Ahmad, E. Choi, N. Jacobs, C. Holmes, M. Mohr, R. Dodhia, J. M. L. Ferres, and J. Marcus, “Fields of The World: A machine learning benchmark dataset for global agricultural field boundary segmentation,” in *Association for the Advancement of Artificial Intelligence (AAAI)*, vol. 2409.16252, Feb. 2025. arXiv: [2409.16252 \[cs.CV\]](#).
- [18] A. Sarkar, A. DiChristofano, S. Das, P. Fowler, N. Jacobs, and Y. Vorobeychik, “Active geospatial search for efficient tenant eviction outreach,” in *Association for the Advancement of Artificial Intelligence (AAAI)*, Feb. 2025. arXiv: [2412.17854 \[cs.LG\]](#).
- [19] S. Sastry, S. Khanal, A. Dhakal, A. Ahmad, and N. Jacobs, “TaxaBind: A unified embedding space for ecological applications,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Feb. 2025. arXiv: [2411.00683 \[cs.CV\]](#).
- [20] M. Lanier, Y. Xu, N. Jacobs, C. Zhang, and Y. Vorobeychik, “Learning interpretable policies in hindsight-observable POMDPs through partially supervised reinforcement learning,” in *IEEE International Conference on Machine Learning and Applications*, Dec. 2024. arXiv: [2402.09290 \[cs.LG\]](#).
- [21] A. Sarkar, S. Sastry, A. Pirinen, C. Zhang, N. Jacobs, and Y. Vorobeychik, “GOMAA-Geo: Goal modality agnostic active geo-localization,” in *Neural Information Processing Systems (NeurIPS)*, Dec. 2024. arXiv: [2406.01917 \[cs.CV\]](#).
- [22] S. Khanal, E. Xing, S. Sastry, A. Dhakal, Z. Xiong, A. Ahmad, and N. Jacobs, “PSM: Learning probabilistic embeddings for multi-scale zero-shot soundscape mapping,” in *ACM Multimedia*, Oct. 2024. DOI: [10.1145/3664647.3681620](#) arXiv: [2408.07050 \[cs.CV\]](#).
- [23] O. Skean, A. Dhakal, N. Jacobs, and L. G. S. Giraldo, “FroSSL: Frobenius norm minimization for self-supervised learning,” in *European Conference on Computer Vision (ECCV)*, Oct. 2024. arXiv: [2310.02903 \[cs.LG\]](#).
- [24] A. Sarkar, A. DiChristofano, S. Das, P. Fowler, N. Jacobs, and Y. Vorobeychik, “Geospatial active search for preventing evictions,” in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, May 2024.
- [25] C. Greenwell, M. Leotta, J. Crall, N. Jacobs, M. Purri, K. Dana, A. Hadzic, and S. Workman, “Watch: Wide-area terrestrial change hypercube,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.
- [26] A. Sarkar, M. Lanier, S. Alfeld, J. Feng, R. Garnett, N. Jacobs, and Y. Vorobeychik, “A visual active search framework for geospatial exploration,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.
- [27] S. Sastry, S. Khanal, A. Dhakal, D. Huang, and N. Jacobs, “BirdSat: Cross-view contrastive masked autoencoders for bird species classification and mapping,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.

- [28] M. Shugaev, I. Semenov, K. Ashley, M. Klaczynski, N. Cuntoor, M. W. Lee, and N. Jacobs, “ArcGeo: Localizing limited field-of-view images using cross-view matching,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2024.
- [29] A. Sarkar, N. Jacobs, and Y. Vorobeychik, “A partially-supervised reinforcement learning framework for visual active search,” in *Neural Information Processing Systems (NeurIPS)*, Dec. 2023. arXiv: [2310.09689](https://arxiv.org/abs/2310.09689).
- [30] S. Khanal, S. Sastry, A. Dhakal, and N. Jacobs, “Learning tri-modal embeddings for zero-shot soundscape mapping,” in *British Machine Vision Conference (BMVC)*, Nov. 2023. arXiv: [2309.10667](https://arxiv.org/abs/2309.10667) [cs.CV].
- [31] Z. Xiong, F. Qiao, Y. Zhang, and N. Jacobs, “StereoFlowGAN: Co-training for stereo and flow with unsupervised domain adaptation,” in *British Machine Vision Conference (BMVC)*, Nov. 2023. arXiv: [2309.01842](https://arxiv.org/abs/2309.01842).
- [32] X. Xing, C. Peng, Y. Zhang, A.-L. Lin, and N. Jacobs, “AssocFormer: Association transformer for multi-label classification,” in *British Machine Vision Conference (BMVC)*, Nov. 2022.
- [33] E. Xing, X. Xing, L. Liu, N. Jacobs, Y. Qu, and G. Liang, “Neural network decision-making criteria consistency analysis via inputs sensitivity,” in *International Conference on Pattern Recognition (ICPR 2022)*, Aug. 2022. DOI: [10.1109/ICPR56361.2022.9956394](https://doi.org/10.1109/ICPR56361.2022.9956394)
- [34] S. Workman, M. U. Rafique, H. Blanton, and N. Jacobs, “Revisiting near/remote sensing with geospatial attention,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 25.33%, Jun. 2022. DOI: [10.1109/CVPR52688.2022.00182](https://doi.org/10.1109/CVPR52688.2022.00182)
- [35] X. Xing, G. Liang, Y. Zhang, S. Khanal, A.-L. Lin, and N. Jacobs, “ADViT: Vision transformer on multi-modality pet images for alzheimer disease diagnosis,” in *IEEE International Symposium on Biomedical Imaging (ISBI)*, Mar. 2022. DOI: [10.1109/ISBI52829.2022.9761584](https://doi.org/10.1109/ISBI52829.2022.9761584)
- [36] H. Blanton, S. Workman, and N. Jacobs, “A structure-aware method for direct pose estimation,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2022. DOI: [10.1109/WACV51458.2022.00028](https://doi.org/10.1109/WACV51458.2022.00028) arXiv: [2012.12360](https://arxiv.org/abs/2012.12360) [cs.CV].
- [37] Y. Zhang, G. Liang, and N. Jacobs, “Dynamic feature alignment for semi-supervised domain adaptation,” in *British Machine Vision Conference (BMVC)*, Nov. 2021. arXiv: [2110.09641](https://arxiv.org/abs/2110.09641).
- [38] G. Liang, X. Xing, L. Liu, Y. Zhang, Q. Ying, A.-L. Lin, and N. Jacobs, “Alzheimer’s disease classification using 2d convolutional neural networks,” in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2021. DOI: [10.1109/EMBC46164.2021.9629587](https://doi.org/10.1109/EMBC46164.2021.9629587)
- [39] Q. Ying, X. Xing, L. Liu, A.-L. Lin, N. Jacobs, and G. Liang, “Multi-modal data analysis for Alzheimer’s disease diagnosis: An ensemble model using imagery and genetic features,” in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2021. DOI: [10.1109/EMBC46164.2021.9630174](https://doi.org/10.1109/EMBC46164.2021.9630174)
- [40] Y. Zhang, G. Liang, Y. Su, and N. Jacobs, “Multi-branch attention networks for classifying galaxy clusters,” in *International Conference on Pattern Recognition (ICPR 2020)*, Acceptance rate: 28.47%, Jan. 2021. DOI: [10.1109/ICPR48806.2021.9412498](https://doi.org/10.1109/ICPR48806.2021.9412498)
- [41] G. Liang, Y. Zhang, X. Wang, and N. Jacobs, “Improved trainable calibration method for neural networks,” in *British Machine Vision Conference (BMVC)*, Sep. 2020.
- [42] M. U. Rafique, H. Blanton, N. Snively, and N. Jacobs, “Generative Appearance Flow: A hybrid approach for outdoor view synthesis,” in *British Machine Vision Conference (BMVC)*, Sep. 2020.
- [43] G. Liang, X. Wang, Y. Zhang, and N. Jacobs, “Weakly-supervised self-training for breast cancer localization,” in *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, (oral), Jul. 2020. DOI: [10.1109/EMBC44109.2020.9176617](https://doi.org/10.1109/EMBC44109.2020.9176617)
- [44] T. Salem, S. Workman, and N. Jacobs, “Learning a dynamic map of visual appearance,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 25%, Jun. 2020. DOI: [10.1109/CVPR42600.2020.01245](https://doi.org/10.1109/CVPR42600.2020.01245)

- [45] S. Workman and N. Jacobs, “Dynamic traffic modeling from overhead imagery,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Acceptance rate: 5.7% (oral), Jun. 2020. DOI: [10.1109/CVPR42600.2020.01233](https://doi.org/10.1109/CVPR42600.2020.01233)
- [46] G. Liang, X. Wang, Y. Zhang, X. Xing, H. Blanton, T. Salem, and N. Jacobs, “Joint 2d-3d breast cancer classification,” in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 18% (oral), Nov. 2019. DOI: [10.1109/BIBM47256.2019.8983048](https://doi.org/10.1109/BIBM47256.2019.8983048) arXiv: [2002.12392](https://arxiv.org/abs/2002.12392).
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- [48] G. Liang, S. Fouladvand, J. Zhang, M. A. Brooks, N. Jacobs, and J. Chen, “GANai: Standardizing CT images using generative adversarial network with alternative improvement,” in *IEEE International Conference on Healthcare Informatics (ICHI)*, Jun. 2019. DOI: [10.1109/ICHI.2019.8904763](https://doi.org/10.1109/ICHI.2019.8904763)
- [49] Z. Bessinger and N. Jacobs, “A generative model of worldwide facial appearance,” in *IEEE Winter Conference on Applications of Computer Vision (WACV)*, (oral), Jan. 2019. DOI: [10.1109/WACV.2019.00172](https://doi.org/10.1109/WACV.2019.00172)
- [50] N. Jacobs, A. Kraft, M. U. Rafique, and R. D. Sharma, “A weakly supervised approach for estimating spatial density functions from high-resolution satellite imagery,” in *ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (ACM SIGSPATIAL)*, Acceptance rate: 22.5% (oral), Nov. 2018. DOI: [10.1145/3274895.3274934](https://doi.org/10.1145/3274895.3274934) arXiv: [1810.09528](https://arxiv.org/abs/1810.09528).
- [51] R. P. Mihail and N. Jacobs, “Automatic hand skeletal shape estimation from radiographs,” in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Acceptance rate: 19.6%, Nov. 2018. DOI: [10.1109/BIBM.2018.8621196](https://doi.org/10.1109/BIBM.2018.8621196)
- [52] S. Schuler, M. Zhai, N. Jacobs, and M. Chandraker, “Learning to look around objects for top-view representations of outdoor scenes,” in *European Conference on Computer Vision (ECCV)*, Acceptance rate: 31.8%, Sep. 2018. DOI: [10.1007/978-3-030-01267-0_48](https://doi.org/10.1007/978-3-030-01267-0_48) arXiv: [1803.10870](https://arxiv.org/abs/1803.10870).
- [53] M. Zhai, T. Salem, C. Greenwell, S. Workman, R. Pless, and N. Jacobs, “Learning geo-temporal image features,” in *British Machine Vision Conference (BMVC)*, Acceptance rate: 29.5%, Sep. 2018.
- [54] C. Greenwell, S. Workman, and N. Jacobs, “What goes where: Predicting object distributions from above,” in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Jul. 2018. DOI: [10.1109/IGARSS.2018.8519251](https://doi.org/10.1109/IGARSS.2018.8519251) arXiv: [1808.00995](https://arxiv.org/abs/1808.00995).
- [55] T. Salem, M. Zhai, S. Workman, and N. Jacobs, “A multimodal approach to mapping soundscapes,” in *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Jul. 2018. DOI: [10.1109/IGARSS.2018.8517977](https://doi.org/10.1109/IGARSS.2018.8517977)
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- [59] N. Vo, N. Jacobs, and J. Hays, “Revisiting IM2GPS in the deep learning era,” in *IEEE International Conference on Computer Vision (ICCV)*, Acceptance rate: 28.9%, Oct. 2017. DOI: [10.1109/ICCV.2017.286](https://doi.org/10.1109/ICCV.2017.286)
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Technical Reports

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- [2] N. Jacobs, S. Schuh, and R. Pless, "On unusual pixel shapes and image motion," Computer Science and Engineering, Washington University in St. Louis, MO, USA, Tech. Rep. WUCSE-2009-16, Jun. 2009.

Datasets

- [1] A. Abrams, J. Tucek, J. Little, N. Jacobs, and R. Pless, *LOST: Longterm observation of scenes (with tracks)*, <http://mvrl.github.io/LOST>.
- [2] M. T. Islam, C. Greenwell, and N. Jacobs, *GeoFaces: A large database of geolocated face patches*, <http://mvrl.github.io/GeoFaces>.
- [3] N. Jacobs, R. Pless, A. Abrams, and many others (see website for details), *AMOS: The archive of many outdoor scenes*, <https://mvrl.github.io/AMOS>.
- [4] P. Mihail, S. Workman, Z. Bessinger, and N. Jacobs, *SkyFinder: A large dataset of webcam images annotated with sky regions*, <https://mvrl.github.io/SkyFinder>.
- [5] M. U. Rafique, H. Blanton, and N. Jacobs, *Brooklyn Panorama Synthesis: A large dataset of panoramic images suitable for view synthesis evaluation*. <https://mvrl.github.io/GAF>.
- [6] T. Salem, S. Workman, M. Zhai, and N. Jacobs, *Cross-View Time (CVT)*, <https://mvrl.github.io/CVT>.
- [7] T. Salem, S. Workman, M. Zhai, and N. Jacobs, *Face2Year: A large number of images extracted from highschool yearbooks*, <https://mvrl.github.io/Face2Year>.

- [8] S. Workman and N. Jacobs, *Cross-View ScenicOrNot (CVSoN)*, <https://mvrl.github.io/CVSoN>.
- [9] S. Workman and N. Jacobs, *Crossview USA (CVUSA): A large dataset containing millions of pairs of ground-level and aerial/satellite images from across the United States*. <https://mvrl.github.io/CVUSA>.
- [10] S. Workman, M. Zhai, and N. Jacobs, *Horizon Lines in the Wild (HLW): A large database of images with known horizon-line location*, <http://mvrl.github.io/HLW>.

Patents

- [1] N. Jacobs and S. Workman, *Network architecture for generating a labeled overhead image*, US Patent App. 16/045,606, Jan. 2020.
- [2] J. A. G. Whitney, J. T. Fessler, Z. C. N. Kratzer, N. B. Jacobs, A. M. Whitney, et al., *Method and system for estimating error in predicted distance using RSSI signature*, Jan. 2016.

5 Talks

- 1. “From Social Media to Sanderlings: Multimodal Tools for Understanding and Mapping Biodiversity” (Keynote), Dec 2025, Workshop on Imageomics @ NeurIPS, San Diego, CA
- 2. “Learning to Map Anything, Anywhere, Anytime” (Sanghi College of Engineering Dean’s Special Invited Speaker), Dec 2025, Northern Arizona University, Flagstaff, AZ
- 3. “Learning to Map Anything, Anywhere, Anytime” (Invited Talk), Nov 2025, National Geospatial-Intelligence Agency, St. Louis, MO
- 4. “(Almost) Two Decades of Vision Across altitudes” (Keynote), Oct 2025, Workshop on 3D-VAST From street to space: 3D Vision Across altitudes (ICCV 2025 Workshop), Honolulu, Hawaii
- 5. “World Foundation Models for GeoAI” (Expert Panel), Sep 2025, Trillion Pixel Challenge Workshop, St. Louis, MO
- 6. “Toward Agentic GEOINT” (Lightning Talk), May 2025, GEOINT Symposium, St. Louis, MO
- 7. “Mapping and Multimodal Embeddings for Biodiversity” (Invited Talk), Mar 2025, Beyond Counting Trees (Bezos Earth Fund), Washington, DC
- 8. “Learning to Map Anything, Anywhere, Anytime” (Keynote), Nov 2023, The International Conference on Digital Image Computing: Techniques and Applications (DICTA), Port Macquarie, Australia
- 9. “Learning to Map Anything, Anywhere, Anytime” (Keynote), Oct 2023, ACM Multimedia Workshop (UAVs in Multimedia: Capturing the World from a New Perspective), Ottawa, Canada
- 10. “A Cost-Sensitive Approach To Dimensionality Reduction for Multispectral Imagery”, Jul 2023, International Geoscience and Remote Sensing Symposium, Pasadena, CA
- 11. “Explorations in Self-Supervised Learning for Change Detection”, Jul 2023, International Geoscience and Remote Sensing Symposium, Pasadena, CA
- 12. “Toward Dynamic Multimodal Remote Sensing: From Buildings and Populations to Soundscapes and Aesthetics”, Apr 2023, Living Earth Collaborative, Washington University, St. Louis, MO
- 13. “Domain-Inspired Deep Learning for Computer Vision, Remote Sensing, and Medical Imaging”, Oct 2022, Imaging Science Seminar, Washington University, St. Louis, MO

14. “Computer Vision for Multimodal Remote Sensing”, Aug 2022, WashU Geospatial Working Group Research Workshop, Washington University, St. Louis, MO
15. “A Structure-Aware Method for Direct Pose Estimation”, Jan 2022, IEEE Winter Conference on Applications of Computer Vision (WACV), Waikoloa Village, HI
16. Panelist for “Non-Traditional Careers in Computer Science” Nov 2021, ACM-W, University of Kentucky, Lexington, KY
17. “Mapping the Visual World Using Webcams, Cell Phones, and Satellites”, Oct 2021, Washington University in St. Louis, MO
18. “Learning Geo-Temporal Scene Models from Webcams, Cell Phones, and Satellites” (Keynote), Oct 2021, International Workshop on Distributed Smart Cameras, an ICCV Workshop (virtual)
19. “Mapping the Visual World Using Webcams, Cell Phones, and Satellites”, Dec 2020, University of Campinas, Unicamp, Brazil (virtual)
20. “Exploring the Intersection of Localization, Mapping, and Image Understanding” (Keynote), Aug 2020, ECCV Workshop on Long-Term Visual Localization (virtual)
21. “Deep Convolutional Neural Networks: Foundations to Frontiers (a 2-day short course)”, Mar 2020, Brazilian Space Agency (INPE), Sao Jose dos Campos, Brazil
22. “What, Where, and When: Mapping the World Using Webcams, Cell Phones, and Satellites”, Mar 2020, Brazilian Space Agency (INPE), Sao Jose dos Campos, Brazil
23. “Learning to Map Visual Appearance”, Feb 2020, Keeping Current Seminar, University of Kentucky (Computer Science), Lexington, KY
24. “Learning to Map Visual Appearance”, Jan 2020, Wageningen University, Netherlands
25. “What, Where, and When: Mapping the World Using Webcams, Cell Phones, and Satellites”, Nov 2019, University of Kentucky (Forestry), Lexington, KY
26. “Learning to Map the Visual World”, Jul 2019, Wright State University, Dayton, OH
27. “Understanding Places Using Ground-Level and Overhead Views” (Keynote), May 2019, Kentucky Geological Society (Annual Symposium), Lexington, KY
28. “Understanding Places Using Ground-Level and Overhead Views”, Feb 2019, Notre Dame University, South Bend, IN
29. “A Generative Model of Worldwide Facial Appearance” (Keynote), Jan 2019, Workshop on Demographic Variations in Performance of Biometric Algorithms, Waikoloa Village, HI
30. “A Generative Model of Worldwide Facial Appearance”, Jan 2019, IEEE Winter Conference on Applications of Computer Vision, Waikoloa Village, HI
31. “A Weakly Supervised Approach for Estimating Spatial Density Functions from High-Resolution Satellite Imagery”, Nov 2018, ACM SIGSPATIAL, Seattle, WA
32. “Understanding Places Using Ground-Level and Overhead Views”, Oct 2018, Commonwealth Computational Summit, Lexington, KY
33. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 4)”, Sep 2018, National Academy of Sciences (IC Academic Research Symposium), Washington, DC

34. “Understanding Places Using Ground-Level and Overhead Views”, Aug 2018, Oak Ridge National Lab, Oak Ridge, TN
35. “WhatGoesWhere: Predicting Object Distributions from Above”, Jul 2018, IGARSS, Valencia, Spain
36. “Building World Models for Situated Training and Planning”, May 2018, Air Force Science and Technology 2030 Workshop, Bloomington, IN
37. “Recent Advances in Image Understanding”, May 2018, DASC, Lexington, KY
38. “(Tutorial) Recent Advances in Deep Learning: Fusing Overhead and Ground-Level Views for Remote Sensing”, April 2018, USGIF Annual Symposium, Tampa, FL
39. “Understanding Places Using Ground-Level and Overhead Views”, Feb 2018, CVPR Area Chair Meeting, Toronto, Canada
40. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 3)”, Sep 2017, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
41. “GPU Accelerated Computer Vision, Remote Sensing, and Machine Learning”, Aug 2017, Kentucky Geological Service, Lexington, KY
42. “Fusing Overhead and Ground-Level Imagery to Improve Scene Understanding”, Jul 2017, Planet, San Francisco, CA
43. “Learning about When and Where from Imagery”, Jun 2017, Orbital Insight, Mountain View, CA
44. “(Tutorial) Recent Advances in Deep Learning: Fusing Overhead and Ground-Level Views for Remote Sensing”, Jun 2017, USGIF Annual Symposium, San Antonio, TX
45. “How Computers See People (extended)”, May 2017, CCTS Biomedical Informatics Seminar Series, Lexington, KY
46. “Understanding Places Using Ground-Level and Overhead Views”, May 2017, Midwest Vision Meeting, Chicago, IL
47. “How Computers See People”, Feb 2017, Suds’n’Science Speaker Series, West Sixth Brewing, Lexington, KY
48. “Learning about When and Where from Imagery”, Feb 2017, University of Missouri, Department of Computer Science
49. “Localization, Mapping, and Image Understanding”, Feb 2017, USGIF Machine Learning Symposium
50. “Deep Convolutional Neural Networks: Concepts and Examples (in Computer Vision)”, Nov 2016, University of Kentucky, Society of Industrial and Applied Mathematics
51. “Crossview Convolutional Networks”, Oct 2016, Applied Imagery and Pattern Recognition, Washington, D.C.
52. “GeoLookbook: Modeling Worldwide Human Visual Appearance (Year 2)”, Sep 2016, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
53. “Deep Convolutional Neural Networks: Concepts and Examples”, Jul 2016, University of Kentucky: Systems Biology and Omics Integration Seminar
54. “Crossview Methods for Localization and Mapping”, Jun 2016, IEEE CVPR Workshop on “Vision from Satellite to Street” (invited talk)
55. “A Fast Method for Estimating Transient Scene Properties”, Mar 2016, Winter Conference on Applications of Computer Vision, Lake Placid, NY

56. “Novel Cues for Geocalibration”, Feb 2016, Indiana University, Bloomington, IN
57. “Novel Cues for Camera Geocalibration”, Jan 2016, Uber Advanced Technology Center, Pittsburgh, PA
58. “Novel Cues for Geocalibration: Cloudy Days, Rainbows, and More”, Oct 2015, Carnegie Mellon University, Pittsburgh, PA
59. “Using Geotagged Internet Imagery to Understand the World”, Sep 2015, Université Laval, Quebec City, Canada
60. “face2gps: Estimating Geographic Location from Facial Features”, Sep 2015, International Conference on Image Processing, Quebec City, Canada
61. “GeoLookbook: Modeling Worldwide Human Visual Appearance”, Sep 2015, National Academy of Sciences (IC Academic Research Symposium), Washington, DC
62. “Exploring the Geo-Dependence of Human Face Appearance”, Mar 2014, Winter Conference on Applications of Computer Vision, Steamboat Springs, CO
63. “Estimating Cloudmaps from Outdoor Image Sequences”, Mar 2014, Winter Conference on Applications of Computer Vision, Steamboat Springs, CO
64. “Scene Geometry from Several Partly Cloudy Days”, Oct 2013, International Conference on Distributed Smart Cameras, Palm Springs, CA
65. “Unlocking the Potential of the Global Network of Outdoor Webcams”, Apr 2013, Rochester Institute of Technology
66. “Geo-temporal Computer Vision: Applications to the NGA”, Nov 2011, National Geospatial-Intelligence Agency
67. “Geo-temporal Computer Vision: Applications to the Army”, Oct 2011, Army Research Lab
68. “Localizing, Calibrating, and Using Thousands of Outdoor Webcams”, Feb 2011, University of North Carolina–Charlotte
69. “Using Clouds Shadows to Infer Scene Structure and Camera Calibration”, Jun 2010, CVPR, San Francisco, CA
70. “Passive Vision and The Power of Collective Imaging”, Apr 2010, Object Video Inc., Reston, VA
71. “Localizing, Calibrating, and Using Thousands of Outdoor Webcams”, Apr 2010, University of Kentucky
72. “Time-Lapse Vision: Localizing, Calibrating, and Using Thousands Outdoor Webcams”, Apr 2010, Google, Mountain View, CA
73. “Passive Vision and The Power of Collective Imaging”, Jan 2010, Google, Mountain View, CA
74. “Incorporating Domain Constraints in Urban Vehicle Tracking”, Nov 2010, University of Missouri, Columbia, MO
75. “Compressive Sensing and Differential Image-Motion Estimation”, Mar 2010, ICASSP, Dallas, TX
76. “The Global Network of Outdoor Webcams: Properties and Applications ”, Nov 2009, ACM GIS, Seattle, WA
77. “Passive Vision: The Global Webcam Imaging Network”, Oct 2009, AIPR, Washington, DC
78. “Calibrating and Using the Global Network of Outdoor Webcams”, Aug 2009, ICDSC, Italy
79. “Adventures in Archiving and Using Three Years of Webcam Images”, Jun 2009, CVPR Workshop on Internet Vision, Miami, FL

80. “Recent Work: Webcams and Grooves”, Aug 2009, Object Video, Reston, VA
81. “Location-Specific Models for Tracking”, Jan 2008, WMVC, Copper Mountain, CO
82. “Using natural cues to geo-locate and geo-orient distributed cameras”, Jan 2008, VISN, Copper Mountain, CO
83. “Foreground Modeling: The Shape of Things That Came”, Feb 2007, WMVC, Austin, Texas

6 Service

University Service

- Washington University in St. Louis (2022–present)
 - 2022–present: Co-Director, Geospatial Research Initiative
 - 2022–present: Research Council, Taylor Geospatial Institute
 - 2023–present: Director of PhD Admissions, Department of Computer Science & Engineering
 - 2023–present: Faculty Technology Advisory Committee, James McKelvey School of Engineering
 - 2023–present: Curriculum Committee, Imaging Science Program
 - 2023–present: Faculty Advisor, WashU Robomaster Club
 - F2024: Faculty Search Committee, Imaging Science Program
 - F2023, F2024: Faculty Search Committee, Computer Science & Engineering Department
 - 2022–2023: Leadership Team, Geospatial Working Group
 - 2022–2023: Strategic Planning Steering Committee, James McKelvey School of Engineering
- University of Kentucky (2010–2022)
 - 2021–2022: Institute for Biomedical Informatics: Steering Committee
 - 2019–2022: Computer Science Department: Executive Committee
 - 2019–2022: College of Engineering: Master Planning/Space Committee
 - 2018–2019, 2020–2022: College of Engineering: Research Advisory Committee
 - 2020–2022: College of Engineering: Graduate Studies Team
 - 2013–2017, 2018–2022: Computer Science Department: Faculty Search Committee
 - 2020–2021: Computer Science Department: Chair Search Committee
 - 2020: College of Engineering: Recruiting Advisory Committee
 - 2018–2019: University Senate (Academic Facilities Committee, Technology Committee)
 - 2017: Member (Information Technology Task Force for Research Enablement and Outreach)
 - 2015–2016: Computer Science Department: ABET Committee
 - 2010–2012, 2015–2016: Computer Science Department: Media and Outreach
 - 2013: Center for Visualization and Virtual Environment: Director Search Committee
 - 2013: Computer Science Department: Chair Search Committee
 - 2012–2013: Computer Science Department: Curriculum Development Committee
 - 2012–2013, 2015: Engineering Day (oral presentation and/or software demonstration)

Professional Service

- Area Chair:
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2018, 2019, 2021, 2023, 2024 (senior), 2025 (lead), 2026 (senior)]
 - European Conference on Computer Vision (ECCV) [2022, 2024 (lead)]
 - IEEE/CVF International Conference on Computer Vision (ICCV) [2023, 2025 (lead)]
 - Conference on Neural Information Processing Systems (NeurIPS) [2024]
 - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2014, 2022, 2023, 2027 (senior)]
- Organizing Committees:
 - IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2019–2023, 2025–2026]
 - Foundational Models Beyond the Visual Spectrum (workshop at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)) [2026]
 - Doctoral Consortium Co-Chair: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2017, 2019, 2024, 2025]
 - Industrial/Government Relations Chair: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2024]
 - Doctoral Consortium Chair: IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2018, 2022]
 - Video Proceedings Chair: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2015]
 - IEEE Workshop on Motion and Video Computing (WMVC) [2011]
- Guest Editor:
 - ISPRS Journal of Photogrammetry and Remote Sensing [2024], Special Issue “Vision Language Models for Remote Sensing Analysis and Interpretation”
 - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS) [2021], Special Issue “Integrating User Generated Contents for Remote Sensing Applications”
 - Elsevier Computer Vision and Image Understanding (CVIU) [2019], Special Issue “Computer Vision for Remote Sensing”
- Panelist:
 - Roundtable Discussion at 1st Workshop on Computer Vision for Earth Observation (CV4EO) Applications 2024 (Hosted as part of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024)
- Session Chair:
 - IEEE International Geoscience and Remote Sensing Symposium (IGARSS) [2020,2023]
 - IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2019]
 - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2016, 2019, 2022, 2024]
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2018, 2024, 2025]

- IEEE/ACM International Conference on Distributed Smart Cameras [2013]
- Reviewing for Journals:
 - Proceedings of the National Academy of Sciences [2024]
 - IEEE Transactions on Geoscience and Remote Sensing [2017, 2020]
 - ISPRS Journal of Photogrammetry and Remote Sensing [2019, 2020]
 - IEEE Transactions on Pattern Analysis and Machine Intelligence [2011×2, 2012, 2018, 2019]
 - IEEE Transactions on Multimedia [2011, 2016]
 - Elsevier Computer Vision and Image Understanding [2010, 2013, 2016×2]
 - IEEE Transactions on Computational Imaging [2016]
 - IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS) [2015]
 - Springer Machine Vision and Applications [2014]
 - IEEE Sensors [2014]
 - Elsevier Image and Vision Computing [2013]
 - IEEE Transactions on Circuits and Systems for Video Technology [2007–2011]
 - IEEE Computer Graphics and Applications [2010]
 - IEEE Transactions on Aerospace and Electronic Systems [2010]
 - Elsevier Computers and Electronics in Agriculture [2010]
 - Cartography and Geographic Information Science [2010]
- Program Committee / Reviewer for:
 - Conferences
 - * IACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR) [2026]
 - * International Conference on Learning Representations (ICLR) [2022, 2025]
 - * IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) [2021, 2024, 2025]
 - * IEEE International Geoscience and Remote Sensing Symposium (IGARSS) [2020]
 - * British Machine Vision Conference (BMVC) [2020]
 - * IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) [2006–2017, 2020]
 - * European Conference on Computer Vision (ECCV) [2010, 2014, 2020]
 - * Neural Information Processing Systems (NeurIPS) [2010–2012, 2020]
 - * AAAI Conference on Artificial Intelligence (AAAI) [2020, 2026 (social impact track)]
 - * IEEE/CVF International Conference on Computer Vision (ICCV) [2007, 2009, 2019, 2021]
 - * Asian Conference on Computer Vision (ACCV) [2010, 2016]
 - * IEEE International Conference on Robotics and Automation (ICRA) [2016]
 - * International Conference on Machine Learning (ICML) [2012]
 - * IEEE International Conference on Advanced Video and Signal-Based Surveillance (AVSS) [2010]
 - Workshops
 - * ECCV Workshop TerraBytes — Towards global datasets and models for Earth observation [2026]
 - * ICLR Workshop on Machine Learning for Remote Sensing [2023]
 - * IEEE/ISPRS Workshop on Large Scale Computer Vision for Remote Sensing Imagery (EARTHVISION) [2017, 2019, 2020]
 - * CVPR Workshop on Photogrammetric Computer Vision [2019]

- * CVPR Workshop on DeepGlobe Satellite Challenge [2018]
 - * ACM International Workshop on Geotagging and Its Applications [2013]
 - * ICCV Workshop on Computer Vision for Converging Perspectives [2013]
 - * IEEE Workshop on Applications of Computer Vision [2012–2013]
 - * ECCV Workshop on Visual Analysis and Geo-Localization of Large-Scale Imagery [2012]
 - * ACM Workshop on Geotagging and Its Applications in Multimedia [2012]
 - * IEEE Workshop on Motion and Video Computation [2009–2011]
- Reviewing for Funding Agencies:
 - Panelist for NSF Information and Intelligent Systems Division [2015, 2016, 2017, 2019]
 - Panelist for NSF Division of Industrial Innovation and Partnerships [2016]
 - External reviewer for NSF Information and Intelligent Systems Division [2015]
 - External reviewer for Fonds de recherche du Quebec [2014]
 - Advisory Committee for:
 - 1st Workshop on Computer Vision for Earth Observation (CV4EO) Applications 2024 (Hosted as part of IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024)

Memberships

- Senior Member: Institute of Electrical and Electronics Engineers
- Full Member: British Machine Vision Association and Society for Pattern Recognition
- Affiliate Member: International Association of Pattern Recognition

Graduate Committee Service

Ph.D. Committees

- | | |
|---|--|
| 1. Ju Shen (Electrical Engineering, 2014) | 12. Anthony Rios (Computer Science, 2018) |
| 2. Chenxi Zhang (Computer Science, 2014) | 13. Nkiruka Uzuegbunam (Electrical Engineering, 2018) |
| 3. Mao Ye (Computer Science, 2014) | 14. Jinping Zhuge (Math, 2019) |
| 4. Yan Huang (Computer Science, 2014) | 15. Xinxin Zuo (Computer Science, 2019) |
| 5. Shaoceng Wei (Statistics, 2015) | 16. Sifei Han (Computer Science, 2019) |
| 6. Bo Fu (Computer Science, 2015) | 17. Kyle Helfrich (Math, 2020) |
| 7. Harikrishnan Unnikrishnan (Electrical Engineering, 2015) | 18. Narjes Bozorg (Electrical Engineering, 2020) |
| 8. Hasan Sajid (Electrical Engineering, 2016) | 19. Alireza Shirvani (Computer Science, 2021) |
| 9. Wesley Hough (Computer Science, 2017) | 20. Sajad Javadinasab Hormozabad (Civil Engineering, 2021) |
| 10. Yajie Zhao (Computer Science, 2017) | 21. Chengxi Li (Computer Science, 2022) |
| 11. Po-Chang Su (Electrical Engineering, 2017) | 22. Tarannum Shaila Zaman (Computer Science, 2022) |

23. David Adeniji (Mechanical Engineering, 2022)
24. Arnab Sarkar (Physics, 2022)
25. Ashutosh Timilsina (Computer Science, 2023)
26. Md Selim (Computer Science, 2023)
27. Stephen Parsons (Computer Science, 2023)
28. Minoo Hosseinzadeh (Computer Science, 2024)
29. Ye Htet (Computer Science, 2024)
30. Gustavo Gratacós (Computer Science, 2024)
31. Junlin Wu (Computer Science, 2025)
32. Hao Liu (Computer Science, 2025)
33. Yiwen Ju (Computer Science, 2025)
34. Weijie Gan (Computer Science, 2025)
35. Sheng-Chieh Lin (Physics, 2025)
36. Tri Pham (Computer Science, 2025)
37. Ruiqi Wang (Computer Science, 2026)
38. Nischal Khanal (Imaging Science, TBD)
39. Anindya Sarkar (Computer Science, TBD)
40. Lisa Liao (Computer Science, TBD)
41. Yu Yan (EECE, TBD)
42. Sidrah Liaqat (Electrical Engineering, TBD)
43. Evelyn Yang (Computer Science, TBD)
44. Ben Warner (Computer Science, TBD)
45. Chengsong Huang (Computer Science, TBD)
46. Yuanlong Su (EECE, TBD)
47. Timur Kasimov (Computer Science, TBD)
48. David Sundarsingh (Electrical and Systems Engineering, TBD)

External Ph.D. Committees

1. Ethan Welty (University of Colorado–Boulder, Environmental Studies, 2018)
2. Yannick Hold-Geoffroy (Laval University, Quebec, CA, Computer Science, 2018)
3. Nam Vo (Georgia Institute of Technology, Computer Science, 2019)
4. Shivangi Srivastava (Wageningen University, Netherlands, Computer Science, 2020)
5. Raian Marett (National Institute for Space Research, Geoinformation Science, 2020)
6. Céline Portenier (University of Bern, Computer Science, 2021)
7. Ahmed Nassar (IRISA, Université Bretagne Sud, Vannes, Computer Science, 2021)
8. Eike Jens Hoffmann (Technical University of Munich, Data Science in Earth Observation, 2022)
9. Morris Alper (Tel Aviv University, Computer Science, 2025)
10. Christoph Gerhardt (Technische Universität Ilmenau, Computer Science, TBD)
11. Rachel Badzioch (Notre Dame University, Biological Sciences, TBD)

Master's Committees

1. Edwin Prem Kumar Sathiyamoorthy (Electrical Engineering, 2011)
2. Hasan Sajid (Electrical Engineering, 2014)
3. Sean Karlage (Computer Science, 2016)
4. Stanley Rosenbaum (Computer Science, 2016)
5. DhiShankar Bhattacharya (Computer Science, 2017)
6. Qingguo Xu (Computer Science, 2017)
7. Xiaofei Zhang (Computer Science, 2017)
8. Ryan Zembrodt (Computer Science, 2019)
9. Jonathan Dingess (Computer Science, 2019)
10. Genghis Goodman (Computer Science, 2019)

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| 11. Subash Khanal (Electrical Engineering, 2020) | 26. Joshua Tang (Computer Science, 2024) |
| 12. Yuan Liu (Computer Science, 2022) | 27. Patrick Lynch (Computer Science, 2024) |
| 13. Aiden McIlraith (Computer Science, 2022) | 28. Owen Ma (Computer Science, 2024) |
| 14. Zhou Chu (Computer Science, 2022) | 29. Evin Jaff (Computer Science, 2024) |
| 15. Zihao Zou (Computer Science, 2022) | 30. Yin Li (Computer Science, 2024) |
| 16. Nan Huang (Computer Science, 2023) | 31. Sizhe Zhang (Statistics, 2024) |
| 17. Di Huang (Computer Science, 2023) | 32. Zifan Wang (Computer Science, 2025) |
| 18. Peizhen Tong (Computer Science, 2023) | 33. Yuxuan Yang (Computer Science, 2025) |
| 19. David Sarpong (Computer Science, 2023) | 34. Haris Naveed (Computer Science, chair, 2025) |
| 20. Yihang Xu (Computer Science, 2023) | 35. Kyle Wolford (Computer Science, 2025) |
| 21. Brian Chao (Computer Science, 2023) | 36. Weining Wang (Computer Science, 2026) |
| 22. Chang Ti (Computer Science, 2023) | 37. Stuart Aldrich (Computer Science, 2026) |
| 23. Emma McMillian (Computer Science, 2023) | 38. Aaron Luo (Computer Science, 2026) |
| 24. Fiona Xu (Computer Science, 2023) | 39. Aadarsha Gopala Reddy (Computer Science, 2026) |
| 25. Kyle Montgomery (Computer Science, 2024) | 40. Yiding Yang (Computer Science, 2026) |

7 Teaching

Courses Taught

The following list summarizes the traditional, classroom courses I have taught:

- *Advances in Computer Vision*, CSE 659a/5519, [F2024,F2025], Washington University in St. Louis
- *Computer Vision*, CSE 559a/5509, [S2023, S2024, S2025, S2026], Washington University in St. Louis
- *Machine Learning*, CS 460g, [F2012, F2013, F2014, F2016, F2018, F2019], University of Kentucky
- *Computer Vision*, CS 636, [S2011, S2013, S2017], University of Kentucky
- *Learning-Based Methods for Computer Vision*, CS 585/685, [S2015], University of Kentucky
- *Advanced Topics in Computer Science: Machine Learning*, CS 685, [S2012], University of Kentucky
- *Intermediate Topics in Computer Science: Computational Photography*, CS 585, [F2010, F2011], University of Kentucky
- *Theory of Computation*, CECS 341, [F2002], University of Missouri

8 Mentoring

Postdoctoral Scholars

- | | |
|---|------------------|
| 1. Benjamin Brodie (Ph.D. Mathematics, University of Kentucky) | Dates: 2020–2022 |
| Research Focus: Change Detection, Object Tracking, Re-Identification, Metric Learning | |

2. [Adeel Ahmad](#) (Ph.D. Geomatics, University of Punjab–Lahore) Dates: 2023–2025
Research Focus: Remote Sensing, Deep Learning, Land-Use Modeling
3. [Subash Khanal](#) Dates: 2025–present
Research Focus: Geospatial Representation Learning, Remote Sensing

Research Staff

1. [Jonathan Lin](#) (MS, Computer Science, Washington University in St. Louis) Dates: 2024–present
Role: Research Associate
2. [Hamza Iqbal](#) (MS, Computer Science, Washington University in St. Louis) Dates: 2026–present
Role: Research Associate

Ph.D. Students

1. [Paul Mihail](#) [*co-chair* with Judy Goldsmith] Degree: Ph.D., Computer Science
Title: Visualizing and Predicting the Effects of Rheumatoid Arthritis on Hands Date: May 2014
Employment: Professor, Computer Science, Valdosta State University
2. [Mohammad T. Islam](#) Degree: Ph.D., Computer Science
Title: Analyzing the Geo-Dependence of Human Face Appearance and Its Applications Date: Jul 2016
Employment: Associate Professor, Computer Science, Southern Connecticut State University
3. [Hamid Hamraz](#) Degree: Ph.D., Computer Science
Title: Computational Forest Modeling using Airborne Remote Sensing LiDAR Date: Apr 2018
Employment: Computational and Data Scientist, Microsoft
4. [Scott Workman](#) Degree: Ph.D., Computer Science
Title: Leveraging Overhead Imagery for Localization, Mapping, and Understanding Date: Apr 2018
Employment: Research Scientist, DZYNE Technologies
5. [Menghua “Ted” Zhai](#) Degree: Ph.D., Computer Science
Title: Deep Probabilistic Models for Camera Geo-Calibration Date: Dec 2018
Employment: Computer Vision Engineer, MatrixTime (startup)
6. [Zach Bessinger](#) Degree: Ph.D., Computer Science
Title: Modeling and Mapping Location-Dependent Human Appearance Date: Dec 2018
Employment: Senior Applied Scientist, Zillow
7. [Tawfiq Salem](#) Degree: Ph.D., Computer Science
Title: Learning to Map the Visual and Auditory World Date: Jul 2019
Employment: Visiting Assistant Professor, Computer and Information Technology, Purdue University
8. [Gongbo “Tony” Liang](#) Degree: Ph.D., Computer Science
Title: Clinical-Inspired Multi-Modal Deep Learning Medical Imaging Analysis Date: Oct 2020
Employment: Assistant Professor, Computer Science, Eastern Kentucky University
9. [Usman Rafique](#) [*co-chair* with Samson Cheung] Degree: Ph.D., Electrical Engineering
Title: Weakly Supervised Learning for Multi-Image Synthesis Date: Jul 2021
Employment: Research Scientist, Kitware Inc.
10. [Hunter Blanton](#) Degree: Ph.D., Computer Science
Title: Revisiting Absolute Pose Regression Date: Aug 2021
Employment: Senior Computer Vision Engineer, Yembo (startup)

11. [Connor Greenwell](#) Degree: Ph.D., Computer Science
 Title: Probabilistic Cross-Domain Representation Learning Date: Jun 2022
 Employment: Senior R&D Engineer, Kitware Inc.
12. [Yu Zhang](#) Degree: Ph.D., Computer Science
 Title: Multimodal Domain Generalization Date: Mar 2023
 Employment: Assistant Professor, Computer Science, Boise State University
13. [Xin Xing](#) [*co-chair* with Ai-Ling Lin] Degree: Ph.D., Computer Science
 Title: Structured Attention for Image Analysis Date: Nov 2023
 Employment: Assistant Professor, University of Nebraska-Omaha
14. [Subash Khanal](#) Degree: Ph.D., Computer Science
 Title: Multimodal Representation Learning for Geospatial Soundscape Mapping Date: Jul 2025
 Employment: Postdoctoral Scholar, Washington University in St. Louis
15. Lydia Reader [*co-chair* with Ross Hammond] Degree: Ph.D., Computational & Data Sciences
 Title: Toward an Understanding of Heterogeneous Effects of Walkability on Children's Physical Activity Date:
 Aug 2026 (est)
16. [Aayush Dhakal](#) Degree: Ph.D., Computer Science
 Title: TBD Date: Dec 2026 (est)
17. [Srikumar Sastry](#) Degree: Ph.D., Imaging Science
 Title: Task-Aligned Multimodal Representation Learning Date: Dec 2026 (est)
18. Michael Lanier [*co-chair* with Yevgeniy Vorobeychik] Degree: Ph.D., Computer Science
 Title: TBD Date: May 2027 (est)
19. [Zhexiao Xiong](#) Degree: Ph.D., Computer Science
 Title: TBD Date: May 2027 (est)
20. [Alex Wollam](#) Degree: Ph.D., Computer Science
 Title: TBD Date: TBD
21. [Nia Hodges](#) Degree: Ph.D., Systems Science and Mathematics
 Title: TBD Date: TBD
22. Tong Li [*co-chair* with Joshua Oltmanns] Degree: Ph.D., Computational & Data Sciences
 Title: TBD Date: TBD
23. [Eric Xing](#) Degree: Ph.D., Computer Science
 Title: TBD Date: TBD
24. [Daniel Cher](#) Degree: Ph.D., Computational & Data Sciences
 Title: TBD Date: TBD
25. [Carlos Lopez de la Cerdá Bazaldu](#) [*co-chair* with Jacob Montgomery] Degree: Ph.D., Computational & Data
 Sciences
 Title: TBD Date: TBD
26. [Feng Qiao](#) Degree: Ph.D., Computer Science
 Title: TBD Date: TBD
27. [Tianyi Gao](#) Degree: Ph.D., Computer Science
 Title: TBD Date: TBD

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| 28. Danni Beaulieu
Title: TBD | Degree: Ph.D., Computer Science
Date: TBD |
| 29. Brian Wei
Title: TBD | Degree: Ph.D., Computer Science
Date: TBD |
| 30. Chih-Chuan (Leo) Hsu
Title: TBD | Degree: Ph.D., Computer Science
Date: TBD |

Visiting Scholars

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| 1. Patrick Tutzauer (University of Stuttgart)
Topic: Geospatial Trajectory Modeling | Dates: Fall 2017 |
| 2. Raian Vargas Maretto (INPE, Brazil)
Topic: Deforestation Detection | Dates: 2018–2019 |
| 3. Rafael Padilha (UNICAMP, Brazil)
Topic: Image Forensics | Dates: 2019–2020 |
| 4. Alex Levering (Wageningen University)
Topic: Landscape Quality Assessment | Dates: 2022–2023 |
| 5. José Nascimento (UNICAMP, Brazil)
Topic: TBD | Dates: 2024 |
| 6. Clemens Mosig (Leipzig University)
Topic: Deep Learning for Tree Mortality Detection | Dates: 2025 |
| 7. Abby Stylianou (Saint Louis University) [<i>Visiting Associate Professor</i>]
Topic: TBD | Dates: 2025–2026 |

Master's Students

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| 1. Feiyu Shi
Title: Principal Component Analysis For Multi-size Images | Degree: MS, Electrical Engineering
Date: Dec 2013 |
| 2. Ryan Baltenberger
Title: Estimating Transient Scene Attributes Using Deep Convolutional Neural Networks | Degree: MS, Computer Science
Date: May 2016 |
| 3. William “Derek” Jones [<i>co-chair</i> with Sally Ellingson]
Title: Scalable Feature Selection and Extraction with Applications in Kinase Polypharmacology | Degree: MS, Computer Science
Date: May 2018 |
| 4. Weilian “William” Song
Title: Image-Based Roadway Assessment using Convolutional Neural Networks | Degree: MS, Computer Science
Date: May 2019 |
| 5. Armin Hadzic
Title: Estimating Free-Flow Speed with LiDAR and Overhead Imagery | Degree: MS, Computer Science
Date: May 2020 |
| 6. David T. Jones
Title: Intensity Harmonization for Airborne LiDAR | Degree: MS, Computer Science
Date: May 2021 |
| 7. Jacob Birge
Title: A Cost-Sensitive Approach To Multimodal Fusion | Degree: MS, Computer Science
Date: Dec 2021 |

8. Zachary Pulliam Degree: MS, Data Science
Title: TBD Date: 2022
9. Alex Greene Degree: MS, Computer Science
Title: Using Aerial Imagery to Estimate Ground-level Object Distributions Date: May 2023
10. Alex Wollam Degree: MS, Computer Science
Title: Exploring Sequential Outdoor Panorama Synthesis with Diffusion Models Date: Jul 2023
11. Hongzhang Wang Degree: MS, Computer Science
Title: Monocular Depth Estimation Date: Dec 2023
12. Nia Hodges Degree: MS, Engineering Data Analytics & Statistics
Title: Wide-Area Image Localization Date: May 2024
13. Wanzhou Liu Degree: MS, Computer Science
Title: High Efficiency Generalizable Driving World Model Date: Fall 2024
14. Lunchi Guo Degree: MS, Computer Engineering
Title: Deep Learning-Based Coordinate Prediction from Medical Fluoroscopic Images for Improved Radiation
Date: Fall 2024
15. Lunchi Guo Degree: MS, Computer Engineering
Title: MedVid-Align: Intelligent Analysis of Medical Procedure Videos Date: Spring 2025
16. Myan Sudharsanan Degree: MS, Computer Engineering
Title: VisionLLMs in the Automotive Domain: Finegrained Details of Traffic Signs and Passenger Vehicles
Date: Spring 2025
17. Ethan Weilheimer Degree: MS (thesis), Computer Science
Title: Partially Supervised Reinforcement Learning for GPS-Denied Navigation Date: Spring 2025
18. Vinh Pham Degree: MS, Engineering Data Analytics & Statistics
Title: Automated Satellite Imagery Analysis for Global Agricultural Field Boundary Detection Date: Spring 2025
19. Shuhan (Steven) Zhang Degree: MS, Computer Science
Title: Query-Specific Feature Transformation for Fine-Grained Image Retrieval Date: Spring 2025
20. Jingyun Ma Degree: MS, Computer Science
Title: Large Language Models for MCNP Input Generation Date: Spring 2025
21. Jackson McCall Degree: MS, Computer Science
Title: GeoSynth++: Large-Scale Satellite Image Synthesis Date: Spring 2025
22. Haris Naveed Degree: MS, Computer Science
Title: Deep Learning for Tree Canopy Height Estimation Date: Spring 2025
23. Rukai Gao Degree: MS, Computer Science
Title: 3D Laryngeal Tumor Mapping via Near-Field SLAM Date: Fall 2026
24. Mohammad Rouie Miab Degree: MS, Computer Science
Title: Improving Semantic Precision in Text-to-Image Diffusion Models via Latent-Space Optimization and Semantically-Parsed Evaluation Date: Spring 2026

Undergraduate Research Students

1. Jim Knochelmann
Title: User-Tools for Aerial Image Registration
Dates: 2011–2012
2. Kyle Kolpek
Title: Aerial Image Registration
Dates: 2012
3. Noora Aljabi
Title: Using Flickr to Map Phenological Trends
Dates: 2013
4. J. David Smith
Title: User-in-the-loop Camera Calibration
Dates: 2013–2015
5. Angelo Stekardis
Title: Understanding Facial Expressions
Dates: 2014–2015
6. Ryan Baltenberger
Title: Understanding Outdoor Scene Appearance
Dates: 2012–2015
7. Connor Greenwell
Title: Interactive Methods for Aerial Imagery Understanding
Dates: 2014–2016
8. Sam Davidson
Title: Applications of Generative Adversarial Networks to Social Media Imagery
Dates: 2016–2017
9. [Aaron Mueller](#)
Title: Deep Learning for Educational Data
Dates: 2018
10. Weilian Song
Title: Applications of Deep Convolutional Neural Networks to Geometric Computer Vision
Dates: 2016–2019
11. Yuhan Long
Title: Deep Learning for Medical Imaging
Dates: 2019
12. Thomas Barber
Title: Deep Learning for Remote Sensing
Dates: 2019
13. Evan Yang
Title: TBD
Dates: 2019
14. [Sean Grate](#)
Title: Deep Learning for Point Clouds
Dates: 2019–2020
15. Nicole Wong
Title: TBD
Dates: 2021
16. [Shashank Bhatt](#)
Title: Multi-Object Tracking
Dates: 2019–2022
17. Evan Bolton
Title: Generating Synthetic Training Data using a Game Engine
Dates: 2021
18. [Julia Stekardis](#)
Title: Large-Scale Image Geo-Localization
Dates: 2021–2022
19. Gareth Walker
Title: Remote Sensing for Social Good
Dates: 2022

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| 20. Matthew Mitchell | Dates: 2022 |
| Title: Remote Sensing for Social Good | |
| 21. Cohen Archbold | Dates: 2020–2022 |
| Title: Automatic Real-Estate Price Estimation | |
| 22. Josh Liu | Dates: 2023 |
| Title: TBD | |
| 23. Sam Schwartz | Dates: 2023 |
| Title: TBD | |
| 24. Daniel Khen | Dates: 2024 |
| Title: TBD | |
| 25. Eddie Choi | Dates: 2024 (Post-Bacc Research Assistant) |
| Title: TBD | |
| 26. John Lim | Dates: 2024 (REU, Northwestern University) |
| Title: TBD | |
| 27. Gabriella Ramirez | Dates: 2024 (REU, University of Missouri) |
| Title: TBD | |
| 28. Brian Song | Dates: 2024 (REU, Vanderbilt University) |
| Title: TBD | |
| 29. Brian Wei | Dates: Fall 2025 |
| Title: Stereo Video Analysis | |
| 30. Dev Gupta | Dates: Fall 2025 |
| Title: Temporally-Consistent Stereo Video Generation | |
| 31. Ice Cui | Dates: Fall 2025 |
| Title: Leveraging Diffusion Transformers to Improve GeoSynth Scalability and Cross-Scale Consistency | |
| 32. Nick Haisler | Dates: 2025 (REU, Drake University) |
| Title: TBD | |
| 33. Rowan Sandvig | Dates: Fall 2025 |
| Title: TBD | |
| 34. Phoenix Jarosz | Dates: Fall 2025 |
| Title: TBD | |
| 35. Kexing (Ariana) Li | Dates: Fall 2025 |
| Title: TBD | |

High School Research Students

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|---|------------------|
| 1. Ryan Baltenberger | Dates: 2011–2012 |
| Title: Gesture-Based User Interaction with the Microsoft Kinect | |
| 2. Alex Lucas | Dates: 2013–2014 |
| Title: Evaluation of Automatic Face Detection Methods | |
| 3. Andrew Tapia | Dates: 2014–2015 |
| Title: Estimating Surface Reflectivity | |

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| 4. Andrew Albrecht | Dates: 2016–2017 |
| Title: Mapping Social Media Imagery | |
| 5. C. J. Labianca | Dates: 2016–2017 |
| Title: Evaluation of Optimization Algorithms for Deep Convolutional Neural Networks | |
| 6. Ryan Landry | Dates: 2017–2018 |
| Title: RRADCL: Rapid Roadway Assessment with Deep Convolutional Learning | |
| 7. Cohen Archbold | Dates: 2017–2018 |
| Title: Photo-Geolocation using Convolutional Neural Networks | |
| 8. Whitney Dawson | Dates: 2019 |
| Title: TBD | |
| 9. Nicole Wong | Dates: 2019–2020 |
| Title: Learning-Based View Synthesis | |
| 10. Chris Wang | Dates: 2019–2021 |
| Title: Multimodal Medical Imaging for Alzheimer’s Disease Classification | |
| 11. William Greenlee | Dates: 2021–2022 |
| Title: Deep Learning for Computer Vision | |
| 12. Krishna Bhatraju | Dates: 2021–2022 |
| Title: Deep Motion Estimation | |